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Soil moisture downscaling using machine learning with scale-aware, physically informed feature constraints

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ABSTRACT

Machine learning enables the downscaling of coarse-resolution remotely sensed soil moisture to finer spatial scales, yet most existing methods lack physical constraints and neglect scale-dependent feature effects. The study proposes a physics-guided machine learning framework that embeds scale-aware feature constraints into a random forest model. These constraints, derived from the Disaggregation based on Physical and Theoretical Scale Change algorithm, adaptively adjust the influence of topography and precipitation across spatial scales. The framework was applied to downscale the European Space Agency's Climate Change Initiative soil moisture product (25 km) to 1 km over the Naqu region of the Tibetan Plateau. Validation against in situ measurements and a 1-km benchmark products showed improved accuracy and stronger physical consistency compared with unconstrained models. Sensitivity analyses confirmed the benefits of integrating physical knowledge and revealed scale-dependent trade-offs in predictor influence. The framework is transferable to heterogeneous landscapes and offers a robust paradigm for generating high-resolution soil moisture products to support hydrological modeling, agricultural water management, and climate impact assessment.

1. Introduction

Surface soil moisture is a key variable in the Earth system, influencing the global water and energy cycles, drought dynamics, agricultural productivity, and ecosystem stability (Yu et al., 2023). It mediates key interactions among precipitation, evapotranspiration, and temperature, making it essential for hydrological modeling, climate studies, and land–atmosphere coupling assessments (Cheng and Liu, 2024; Dos Santos Araujo et al., 2024; Seneviratne et al., 2010). With climate change accelerating and extreme weather events increasing, there is an urgent demand for high-resolution, spatiotemporally continuous soil moisture datasets to support precision agriculture, water resource management, and environmental monitoring.

While in situ measurements provide accurate point-level soil moisture observations, their sparse spatial distribution restricts their applicability for large-scale analyses. In contrast, microwave remote sensing offers wide-area soil moisture observations and has led to the development of several global-scale soil moisture products (Mohanty et al., 2017; Xie et al., 2024), including the Advanced Microwave Scanning Radiometer for the Earth Observing System (AMSR-E) (Brocca et al., 2011), Advanced Microwave Scanning Radiometer-2 (AMSR2) (Zhang et al., 2021), Soil Moisture and Ocean Salinity (SMOS) (Wigneron et al., 2021), and European Space Agency Climate Change Initiative (ESA CCI) (Zhao et al., 2021). However, due to sensor design limitations, these products typically exhibit coarse spatial resolutions (25–50 km), which are insufficient for local-scale applications such as farm-level irrigation planning or localized drought detection.

Spatial downscaling techniques address this resolution gap by integrating coarse-scale soil moisture with higher-resolution auxiliary data that capture spatial heterogeneity in soil moisture dynamics (Peng et al.,

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2017). Existing methods span empirical, semi-empirical, and physically based approaches (Meng et al., 2024). Among them, empirical downscaling using machine learning has gained prominence due to its capacity to model complex nonlinear relationships between soil moisture and auxiliary features such as vegetation indices, land surface temperature, soil texture, albedo, and topography (Jin et al., 2021; Liu et al., 2020; Wang et al., 2024; Zhao et al., 2022). However, traditional data-driven approaches often suffer from two major limitations. First, they rely heavily on statistical correlations without explicitly incorporating physical mechanisms, limiting both their interpretability and robustness. Second, they frequently assume that coarse-scale relationships hold at finer scales, neglecting the inherent scale-dependent variability of soil moisture processes (Senanayake et al., 2024). For instance, large-scale soil moisture variability may be driven by climatic factors and land cover, while finer-scale patterns are more influenced by topography and soil properties (Crow et al., 2012). Overlooking such multi-scale interactions can introduce biases and reduce downscaling accuracy.

Recently, physics-guided machine learning has emerged as a promising strategy to combine physical knowledge with data-driven models (Karniadakis et al., 2021; Meng et al., 2025). Three broad strategies have emerged: (i) incorporating physically meaningful features such as the temperature vegetation dryness index (TVDI) or soil evaporative efficiency (SEE) (Chen et al., 2019; Sun and Gao, 2023); (ii) embedding physical constraints into model structures or loss functions (Xu et al., 2024), and (iii) fusing outputs from physical and machine learning models via assimilation or hybrid modeling (Li et al., 2024). While these approaches improve physical consistency, most still assume scale-invariant relationships and rarely address explicit scale-dependent effects during downscaling.

In this study, we propose a soil moisture downscaling framework that integrates physics-constrained features into machine learning models, explicitly accounting for scale-dependent variability. The framework adopts a feature-level integration strategy, inspired by the semi-empirical DISPATCH (Disaggregation based on Physical and Theoretical Scale Change) algorithm (Molero et al., 2016), which uses soil evaporative efficiency as a physically grounded proxy to infer sub-pixel soil moisture variability. Building on this, our framework dynamically adjusts the influence of auxiliary features based on their scale-sensitive relevance. The method involves four steps: (i) establish relationships between coarse-scale soil moisture and auxiliary features using machine learning; (ii) refine these relationships at finer scales by incorporating physics-constrained features, with constraint strength derived from DISPATCH; (iii) apply geostatistical methods to downscale residuals and produce high-resolution residual fields; and (iv) generate fine-scale trend surfaces based on refined relationships and combine them with residuals to produce final soil moisture maps.

We validated the framework by downscaling the 25-km ESA CCI soil moisture product to 1 km over a representative study area. The ESA CCI dataset is widely used in soil moisture downscaling studies due to its long-term consistency, global coverage, and harmonized multi-sensor retrievals. Validation was conducted using in situ measurements and multiple existing high-resolution soil moisture products, and additional comparative experiments and sensitivity analyses were performed to assess model robustness and generalizability. The specific objectives are to: (i) develop a physics-informed feature integration strategy to enhance downscaling accuracy and robustness; (ii) explicitly incorporate scale-dependent variability for improved fine-scale SM representation; and (iii) demonstrate applicability across heterogeneous landscapes for hydrological, agricultural, and climate-related studies.

2. Materials and methods

2.1. Study area

The Tibetan Plateau in southwestern China plays a crucial role in global climate and hydrological systems due to its unique location, complex geology, and diverse climatic and ecological conditions. This study focuses on the Naqu region in the central Tibetan Plateau (30°–36.5°N, 85°–95°E; Fig. 1), characterized by high-altitude plateau and hilly terrains with gently undulating slopes and an average elevation above 4500 m (Song et al., 2022). Located in the transition zone between semi-humid and semi-arid climates of the sub-frigid zone, the area receives about 400 mm annual precipitation (Zhu et al., 2023). Vegetation shifts from alpine grasslands in the northwest to alpine shrub–meadows in the southeast. Fig. 1 shows the 25-km ESA CCI soil moisture product and a 1-km product for 11 July 2020, along with in situ observation sites (data sources in Section 2.2). Topographic variation, vegetation gradients, and hydro-climatic heterogeneity result in substantial spatial variability in soil moisture, posing challenges for conventional downscaling methods. These complex environmental gradients make the Naqu region an ideal testbed for our scale-aware, physics-guided downscaling framework.

2.2. Dataset

2.2.1. Coarse-scale remotely sensed soil moisture data

The coarse-scale remotely sensed soil moisture data used in this study were obtained from the ESA CCI dataset. The ESA CCI soil moisture product provides a standardized, long-term global record by integrating passive and active microwave observations into three categories: passive, active, and combined products. Since its launch in 2010, the dataset has offered surface soil moisture at a spatial resolution of 25 km × 25 km and has been widely applied in hydrological, agricultural, and climate-related research, particularly as a key input for spatial downscaling studies. Different product versions are available, varying in their merging algorithms and input data sources. In this study, we selected the combined soil moisture product (version v07.1) because it leverages the complementary strengths of passive and active microwave sensors, improving spatial coverage and temporal consistency. The dataset was extracted for the period of July–August from 2016 to 2020 over the study region. Further details on retrieval principles, merging algorithms, and quality control procedures are provided in the ESA CCI documentation (https://www.esa-soilmoisture-cci.org/).

2.2.2. Auxiliary feature data

To improve the spatial resolution of the downscaled soil moisture, we incorporated a comprehensive set of auxiliary features spanning vegetation, climate, terrain, soil properties, and physically based soil moisture indices. Vegetation information was represented by the normalized difference vegetation index (NDVI), obtained from the MODIS MOD13A2 product at 1 km spatial resolution and 16-day intervals, reflecting canopy greenness and density. Climate variables included daytime (LST_D) and nighttime (LST_N) land surface temperature from the MODIS MOD11A2 product (1 km, 8-day composite), albedo from the MODIS MCD43A3 daily product (1 km) calculated as a weighted sum of black-sky and white-sky albedo (weights 0.34 and 0.66), and precipitation from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) dataset, which integrates satellite infrared observations with in situ measurements at 0.05° (~5.5 km) resolution; these precipitation data were aggregated into 5-day means for this study. All MODIS and CHIRPS datasets were accessed via the Google Earth Engine platform.

Terrain variables, including elevation, slope, aspect, and the topographic wetness index (TWI), were derived from the Shuttle Radar Topography Mission Digital Elevation Model (SRTM DEM) at 30 m resolution and resampled to 1 km to match other inputs. Soil property data were obtained from the National Tibetan Plateau / Third Pole Environment Data Center and included soil organic carbon, pH value, total nitrogen, total phosphorus, total potassium, cation exchange capacity, sand, silt, clay, soil texture type, unit weight, soil thickness, etc. The original 250 m soil property data were resampled to 1 km for

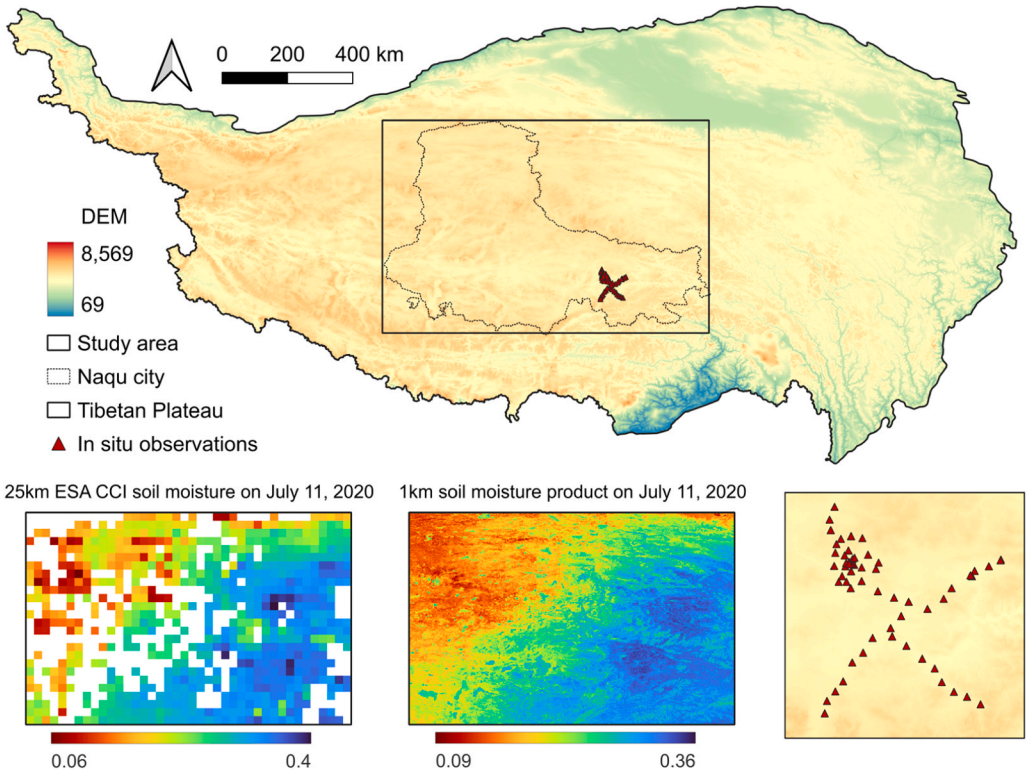


Fig. 1. Location and elevation of the study area in the central Tibetan Plateau. The 25-km ESA CCI soil moisture product and a 1-km soil moisture product for 11 July 2020 are shown, and the distribution of in situ observation sites from the Naqu soil moisture and temperature monitoring network is enlarged.

consistency. In addition, two physically based indices (TVDI and SEE) were derived from remote sensing observations of LST and NDVI, capturing short-term variations in soil moisture driven by water stress and surface energy exchanges.

2.2.3. *In situ observations and 1-km soil moisture products*

Ground truth data were sourced from the Naqu soil moisture and temperature monitoring network. The network comprises 57 automated stations measuring soil moisture at 0–5 cm depth, temperature, and freeze–thaw status, with observation frequencies of 30 min. For this study, daily average soil moisture values from July–August (2016–2020) were calculated and used for validation. The data were acquired from the National Tibetan Plateau / Third Pole Environment Data Center (Chen et al., 2022).

To ensure a comprehensive evaluation of the proposed downscaling framework, three existing 1-km soil moisture products were assembled for comparison (see Table 2). Each product was derived from coarse-resolution satellite soil moisture observations but using distinct downscaling principles, enabling a methodological intercomparison across representative approaches. The BTCH (Bayesian Three-Cornered Hat) 1-km daily soil moisture dataset over the Tibetan Plateau (2001–2020) (Shangguan et al., 2023b) served as the primary benchmark. BTCH was specifically designed for the Tibetan Plateau and is generated through the fusion of predictions from multiple machine-learning models, including artificial neural networks (ANNs), convolutional neural networks (CNNs), residual networks (ResNet), long short-term memory (LSTM) networks, and XGBoost (Shangguan et al., 2023a). All models were trained using ESA CCI soil moisture inputs, and a Bayesian fusion framework was applied to integrate their outputs into a spatially consistent high-resolution product. Because BTCH uses the same input data source (ESA CCI) but relies on fundamentally different downscaling mechanisms, it provides an independent and robust baseline for assessing the performance of our method.

Two nationwide 1-km soil moisture products were also included to support broader methodological comparisons (Shangguan et al., 2024; Song et al., 2025). Following the original publications (Li et al., 2022; P. Song et al., 2022), they are referred to as PM-based SSM (passive-microwave-derived surface soil moisture) and SMCI (Soil Moisture of

Table 1
Data sources, spatial/temporal resolution, and thematic categories of auxiliary features used for soil moisture downscaling.

Category	Products	Feature	Resolution	Source
Vegetation	MODIS/061/MOD13A2	NDVI	1 km/16-day	https://earthengine.google.com
Climate	MODIS/061/MOD11A2	LST_D, LST_N	1 km/8-day	
	MODIS/061/MCD43A3	Albedo	1 km/daily	
	UCSB-CHG/CHIRPS/PENTAD	Precipitation	0.05°/5-day	
Terrain	SRTM-DEM	DEM, Slope, Aspect, TWI	30 m/Static	http://srtm.csi.cgiar.org
Soil properties	Dataset of digital soil mapping products for the Qinghai-Tibet Plateau (2015–2024)	BD, Sand, Silt, Clay, SOC, pH, Thickness, etc.	250 m/ Static	https://doi.org/10.11888/Terre.tpdc.272482
Soil moisture-related indices	Derived from LST & NDVI	TVDI, SEE	1 km/16-day	https://earthengine.google.com

Abbreviations: DEM – digital elevation model; NDVI – normalized difference vegetation index; LST_D and LST_N – daytime and nighttime land surface temperature; TWI – topographic wetness index; BD – soil bulk density; SOC – soil organic carbon; TVDI – temperature vegetation dryness index; SEE – soil evaporative efficiency.

Table 2
Summary of in situ and 1-km soil moisture datasets used for validation and comparison.

Dataset	Spatial Resolution	Spatial Coverage	Source/Method	Study Purpose
In situ observations	Point scale (0–5 cm depth)	Naqu region, Tibetan Plateau	Ground stations	Ground truth for validation
BTCH	1 km	Tibetan Plateau	Ensemble of machine learning models	Primary benchmark for validation
PM-based SSM	1 km	Nationwide (China)	Physically guided downscaling	Cross-validation
SMCI	1 km	Nationwide (China)	Machine learning-based downscaling	Cross-validation

Abbreviations: BTCH – Bayesian Three-Cornered Hat; PM-based SSM – passive-microwave-derived surface soil moisture; SMCI – Soil Moisture of China by in situ data.

China by in situ data). The PM-based SSM dataset employs a physically guided downscaling approach conceptually similar to DISPATCH, representing physics-constrained methods. In contrast, the SMCI product mainly relies on machine-learning-based downscaling and thus reflects data-driven strategies.

2.3. Machine learning-based downscaling method

The proposed downscaling framework employs a two-stage strategy that decomposes the target variable into a deterministic trend component and a stochastic residual component (Jin et al., 2021). Let $\{z(S_i), S_i \in A\}$ and $\{z(s_j), s_j \in A\}$ denote the values of a target variable (e.g., soil moisture) at coarse and fine spatial resolutions, respectively, within a given spatial domain A. Here, $S_i = (U_i, V_i)$ and $s_j = (u_j, v_j)$ represent the geographic coordinates of the pixel centers at coarse and fine scales. The fine-resolution value of the target variable can be expressed as the sum of a trend and a residual term:

$$z(s_j) = m(s_j) + \varepsilon(s_j) \tag{1}$$

where $m(s_j)$ is the trend surface, modeled as a function of auxiliary features, and $\varepsilon(s_j)$ denotes the residual, representing spatial variation not captured by the trend.

***Trend surface modeling*:** To construct the trend surface, a set of p auxiliary variables $\{x_k(S_i), k = 1, \ldots, p, p > 1\}$ is used to model the coarse-scale observations $\{z(S_i)\}$. A random forest regression model is trained to learn the nonlinear mapping:

$$z(S_i) \approx f\left(x_1(S_i), \ldots, x_p(S_i)\right) \tag{2}$$

where $f(\cdot)$ is the fitted random forest model, and $x_k(S_i)$ represents the value of the kth auxiliary feature at the coarse-scale pixel S_i. The residuals at the coarse scale are then computed as:

$$\hat{\varepsilon}(S_i) = z(S_i) - f\left(x_1(S_i), \ldots, x_p(S_i)\right) \tag{3}$$

These residuals capture the part of the coarse-scale soil moisture that is not explained by the auxiliary data.

***Residual downscaling via Kriging*:** In practical applications, owing to spatial heterogeneity, the residual that remains after removing the trend surface retains a certain degree of spatial autocorrelation. The area-to-area kriging (ATAK) method, commonly applied for residual downscaling, is employed to downscale large-scale residuals to the intended scale. ATAK operates on the principle of the Kriging interpolation method, segmenting the spatial region into numerous small faces and performing Kriging interpolation on each face to estimate the entire spatial region. This method utilizes the weighted sum of pixel values at a larger scale to estimate values at a smaller scale. The residuals of a fine-scale pixel s_j can be computed through the weighted sum of residuals of neighboring pixels at the coarse scale S_i, expressed as:

$$\hat{\varepsilon}(s_j) = \sum_{i=1}^{L} \lambda_i(s_j) \bullet \hat{\varepsilon}(S_i) \tag{4}$$

where $\hat{\varepsilon}(s_j)$ denotes the residuals of a fine-scale pixel s_j. L represents the number of selected neighboring pixels, and $\lambda_i(s_j)$ is the weight coefficient of ATAK, satisfying the certain condition given by $\sum_{l=1}^{L} \lambda_i(s_j) = 1$. Following the Kriging principle, these weights are estimated by minimizing the variance of prediction errors while maintaining unbiasedness.

Fine-scale reconstruction: Assuming the learned mapping $f(\cdot)$ holds at finer resolutions, the trend surface at the fine scale can be obtained by evaluating the model with fine-scale auxiliary data:

$$\hat{m}(s_j) = f\left(x_1(s_j), \ldots, x_p(s_j)\right) \tag{5}$$

Finally, the downscaled fine-resolution estimate of the target variable is reconstructed by combining the estimated trend and residual components:

$$\hat{z}(s_j) = \hat{m}(s_j) + \hat{\varepsilon}(s_j) \tag{6}$$

This two-stage approach allows for the integration of auxiliary data through machine learning and captures spatial dependencies via geostatistical interpolation, thereby enhancing the spatial detail and physical consistency of downscaled soil moisture products.

2.4. Machine learning with DISPATCH-constrained features

The assumption of scale-invariant relationships between auxiliary features and the target variable, as implied in Eq. (5), often introduces uncertainty in the trend surface model, particularly when transferring models from coarse to fine spatial resolutions. To address this issue, we integrate the DISPATCH method into the downscaling framework to refine the trend surface using physically guided constraints, thereby improving its transferability and reliability at finer scales.

DISPATCH is a semi-empirical algorithm originally developed for downscaling soil moisture by combining physically meaningful variables and a first-order Taylor series expansion (Merlin et al., 2012). It estimates fine-scale soil moisture ($SM_{dispatch}$) using high-resolution LST and vegetation indices through their influence on SEE, a physically relevant indicator of soil–atmosphere interaction.

The DISPATCH formulation expresses the relationship between coarse-scale and fine-scale soil moisture as:

$$SM_{dispatch} = SM_{CR} + \left(\frac{\partial SM}{\partial SEE}\right)_{CR} \bullet (SEE_{HR} - SEE_{CR}) \tag{7}$$

where SM_{CR} represents the original soil moisture at a coarse scale, and SM_{HR} signifies the downscaled soil moisture at a finer scale. The symbol $\left(\frac{\partial SM}{\partial SEE}\right)_{CR}$ is the partial derivative of soil moisture concerning soil evapotranspiration efficiency. SEE_{HR} represents the fine-scale soil evaporation efficiency, and SEE_{CR} is the spatial average of SEE_{HR} within a particular pixel.

Assuming residuals are small compared to the trend component, the ratio of fine- to coarse-scale trend surface values can be approximated by the ratio of DISPATCH-derived soil moisture to the machine-learning-derived trend:

$$\frac{\hat{m}'(s_j)}{m(s_j)} \approx \frac{SM_{dispatch}}{f\left(x_1(s_j), \ldots, x_p(s_j)\right)} \tag{8}$$

We define this ratio as a constraint factor ω, which is then used to adjust the influence of select auxiliary variables within the trend surface model:

$$\omega \approx \frac{SM_{dispatch}}{f(x_1(s_j), \ldots, x_p(s_j))} \quad (9)$$

Accordingly, the DISPATCH-constrained trend surface at fine scale is computed as:

$$\hat{m}'(s_j) = f(\boldsymbol{x}(s_j), \quad \omega^d \bullet \boldsymbol{x}'(s_j)) \quad (10)$$

where $\boldsymbol{x}(s_j)$ represents the original set of (unconstrained) auxiliary features, and $\boldsymbol{x}'(s_j)$ denotes the subset of auxiliary features to which the physical constraint is applied. The parameter ω adjusts the relative influence of auxiliary features based on physical insights, while $d \geq 1$ controls the strength of the constraint.

In this study, we focus the physical constraint on topographic and precipitation-related features, based on their known scale-dependent impacts on soil moisture dynamics. Specifically:

Topographic features (e.g., elevation, slope, aspect) often exert stronger control at fine spatial scales by influencing microclimate, runoff, and water retention. However, their effects are often underrepresented in coarse-scale models. Hence, for pixels where $\omega > 1$, we amplify the influence of topography in the trend surface modeling.

Precipitation, on the other hand, dominates soil moisture variability at broader scales, reflecting large-scale atmospheric processes. Its predictive power tends to diminish at finer resolutions due to increased heterogeneity in infiltration, drainage, and localized redistribution. Therefore, for pixels where $\omega < 1$, the influence of precipitation is down weighted.

This feature-specific, spatially adaptive weighting mechanism enables the trend surface model to better reflect the underlying physical controls on soil moisture across scales, enhancing both interpretability and predictive performance of the downscaled soil moisture estimates.

2.5. Overview of downscaling strategy

The proposed spatial downscaling framework was implemented to generate 1-km resolution soil moisture products from 25-km ESA CCI soil moisture data over the Naqu region of the Tibetan Plateau. Downscaling was conducted on 20 dates spanning July and August from 2016 to 2020. The resulting high-resolution soil moisture maps were evaluated through comprehensive comparative analyses, using both in situ observations and the 1-km BTCH soil moisture product as reference datasets.

The framework integrates machine learning, physical constraints, and geostatistical interpolation as follows (Fig. 2):

***Auxiliary feature selection*:** multi-scale (1 km, 5 km, 25 km) variables were used, including NDVI, LST, albedo, precipitation, TVDI, SEE, topography (elevation, slope, aspect, TWI), and soil properties (texture, SOC, pH, bulk density, thickness, etc.). Datasets were resampled via spatial averaging or bilinear interpolation.

***Coarse-scale trend surface modeling*:** A random forest regression was trained on ESA CCI soil moisture and 25-km auxiliary features. Hyperparameters (n_estimators, max_depth, min_samples_leaf, max_features) were optimized via Bayesian Optimization with five-fold cross-validation.

***Intermediate-scale modeling and DISPATCH integration*:** To bridge

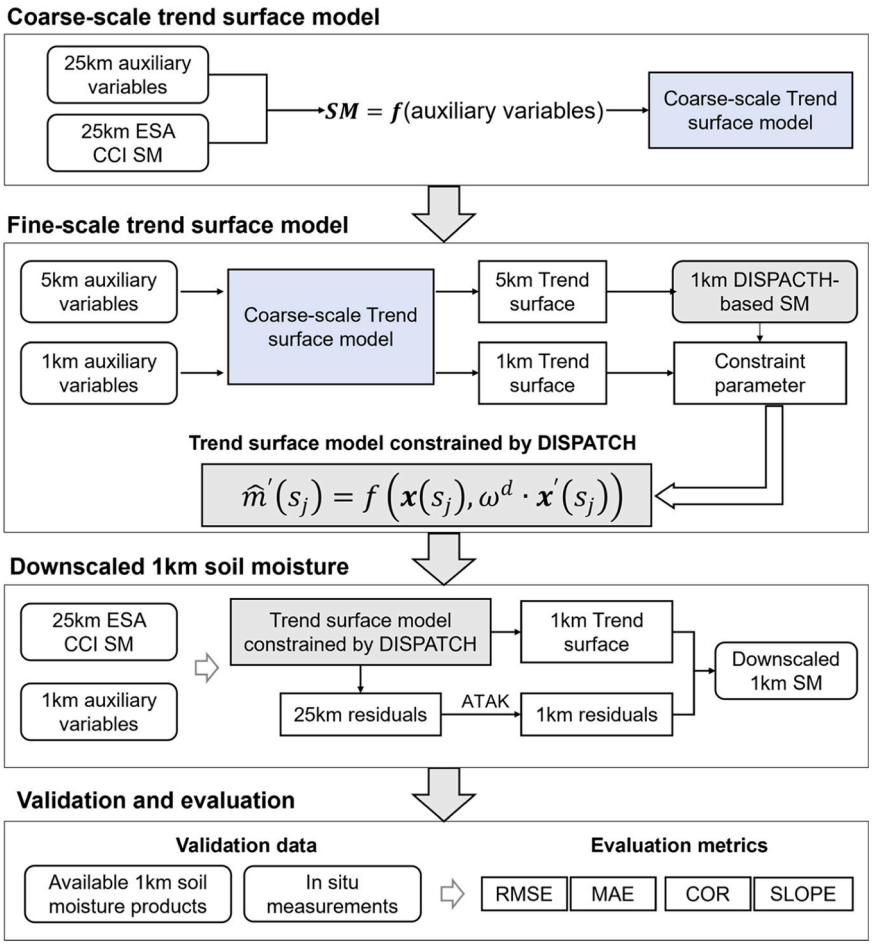


Fig. 2. Framework of the spatial downscaling method applied to 25-km ESA CCI soil moisture data in this study. The process integrates random forest-based trend surface modeling, DISPATCH-constrained feature adjustment, and residual refinement via area-to-area Kriging (ATAK) to produce high-resolution (1-km) soil moisture estimates.

the substantial scale gap between 25 km and 1 km and mitigate artifacts from abrupt disaggregation, an intermediate 5-km trend surface was generated using the coarse-scale random forest model and 5-km auxiliary data. The DISPATCH algorithm was then applied to refine these predictions, producing a 1-km DISPATCH-based soil moisture map. A constraint ratio computed as the ratio between DISPATCH estimates and the initial 1-km trend surface (from the random forest model), was used to scale selected auxiliary features, particularly topographic metrics, within the fine-scale trend model.

***Downscaling prediction and evaluation*:** Residuals from the coarse-scale model (25 km) were spatially interpolated to 1 km using the ATAK method. The refined 1-km trend surface and downscaled residuals were combined (Eq. 1) to produce the final soil moisture product. Performance was evaluated against in situ measurements and the 1-km BTCH product using Root mean square error (RMSE), mean absolute error (MAE), Pearson correlation coefficient (COR), linear regression slope (SLOPE) between predicted and reference values.

2.6. Validation and sensitivity analyses

The performance of the downscaled soil moisture product was evaluated against two reference datasets: (1) in situ soil moisture measurements from the Naqu network, and (2) three 1-km soil moisture products. Accuracy, linearity, and bias were quantified using four metrics: RMSE, MAE, COR, and SLOPE between predicted and reference values. These evaluations served to assess both the predictive skill and the physical consistency of the proposed framework.

To investigate the impact of auxiliary feature composition and the application of physical constraints, we conducted a series of controlled comparative experiments (Table 3). The experimental design was structured to isolate two key factors: (1) the contribution of additional physically meaningful variables, and (2) the effect of DISPATCH-based constraints on selected predictors in the trend surface model. Each configuration was assessed against both in situ observations and the BTCH product. The configurations are as follows:(Table 4)

$Base_1$: A baseline configuration using standard auxiliary features—LST, NDVI, albedo, precipitation, slope, aspect, and elevation—following the downscaling method described in Section 2.3, without any physical feature constraints.

$Base_2$: Builds upon $Base_1$ by incorporating soil property variables.

$Variable_1$: Builds on $Base_1$ by adding physically meaningful auxiliary variables, namely TVDI and SEE, to assess their contribution to the downscaling process.

$Variable_2$: Extends $Base_2$ with the inclusion of TVDI and SEE, providing a framework to examine their effects in conjunction with soil properties.

Table 3
Experimental design for comparison of downscaling methods informed by soil moisture physical features.

Experiment type	Name	Auxiliary features	Feature constraints in trend surface model
Basic experiment	$Base_1$	LST, NDVI, Albedo, Precipitation, Aspect, Slope, DEM	None
	$Base_2$	$Base_1$ + Soil properties	
Physical auxiliary	$Variable_1$	$Base_1$ + TVDI, SEE	
	$Variable_2$	$Base_2$ + TVDI, SEE	
Physical constraint	$Proposed_1$	Same as $Variable_1$	Enhance topographic influence
	$Proposed_2$	Same as $Variable_1$	Enhance topographic influence; reduce precipitation impact
	$Proposed_3$	Same as $Variable_2$	Enhance topographic influence
	$Proposed_4$	Same as $Variable_2$	Enhance topographic influence; reduce precipitation impact

Table 4
Relative change rates (%) in evaluation metrics for 1-km downscaled soil moisture after applying DISPATCH constraints, using in situ observations as reference. Negative values for MAE and RMSE indicate reduced error, while positive values for COR and SLOPE indicate performance improvement. Comparisons are made against both the unconstrained model and the BTCH product.

Year	Change rate (%)				Comparison of evaluation measures
	MAE	RMSE	COR	SLOP	
2016	0.97	0.44	-0.88	-0.92	(Constrained – Unconstrained)
2017	-1.85	-1.54	10.74	12.10	Unconstrained
2018	1.12	0.50	-4.83	-3.01	
2019	-1.18	-1.49	4.58	6.35	
2020	-1.18	-0.9	5.40	6.80	
2016	-5.00	-3.84	-0.56	33.76	(Constrained – BTCH)
2017	-2.91	-0.95	-26.19	56.41	BTCH
2018	-1.59	-2.80	-27.43	19.11	
2019	-6.89	-5.54	6.91	76.83	
2020	2.35	1.45	-24.51	21.01	

$Proposed_1$: Builds on $Variable_1$ by incorporating physical constraints from DISPATCH, specifically adjusting the influence of topographic variables in the trend surface model to enhance their impact.

$Proposed_2$. Further extends $Proposed_1$ by simultaneously reducing the influence of precipitation in the trend surface model, while maintaining the enhanced weighting for topographic variables.

$Proposed_3$. Builds on $Variable_2$ by applying DISPATCH-based constraints to enhance the contribution of topographic features in the trend surface model.

$Proposed_4$. Extends $Proposed_3$ by also reducing the influence of precipitation, aiming to assess the combined effect of constraining both topographic and precipitation variables.

This framework enables a systematic sensitivity analysis of the role of auxiliary data type and constraint strategy in improving the accuracy, robustness, and physical interpretability of the downscaled soil moisture estimates.

3. Results

3.1. Downscaled soil moisture from ESA CCI data

The spatial distribution of downscaled soil moisture revealed notable variations across the study area. Generally, soil moisture in the northwest was lower compared to the southeast, as illustrated by the 1-km downscaled soil moisture maps for five selected dates (Fig. 3). The spatial distribution of soil moisture was influenced by multiple factors and did not show a strong correlation with the DEM. The spatial patterns of soil moisture in 2016 and 2017 were relatively similar, with the southwest and southeast corners exhibiting higher moisture levels. On July 28, 2018, the study area experienced widespread higher soil moisture, except for a localized drought area in the northwest corner. However, by 2019, the drought in the northwest had expanded, resulting in a pronounced accumulation of low soil moisture values. By July 27, 2020, soil moisture levels across the study area generally ranged between 0.2 and 0.3 m^3/m^3, with pronounced spatial heterogeneity. These results highlight the variability and stratification in soil moisture distribution over time.

3.2. Downscaled soil moisture against in situ observations

The proposed downscaling framework outperformed existing 1-km soil moisture products in terms of both accuracy and consistency with in situ measurements (Fig. 4). The accuracy of the downscaled 1-km soil moisture estimates was evaluated against in situ observations collected over 20 time points between July and August from 2016 to 2020. Overall, the downscaled soil moisture showed good agreement with the ground measurements, achieving a MAE of 0.070 m^3/m^3 and a RMSE of

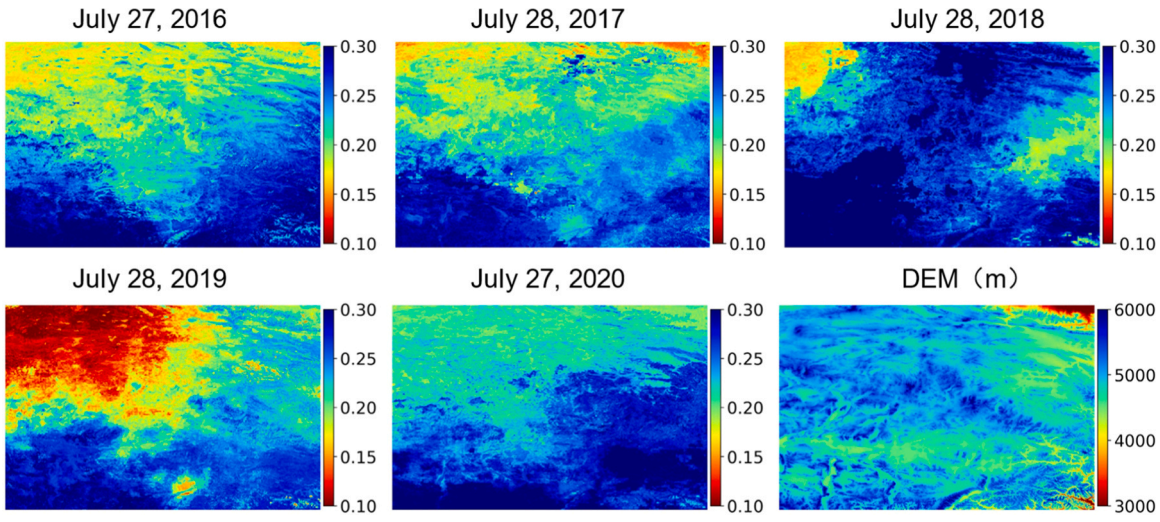


Fig. 3. Downscaled 1-km soil moisture maps for five selected dates between 2016 and 2020, illustrating spatiotemporal variability across the Naqu region. The digital elevation model (DEM) is also shown to provide topographic context.

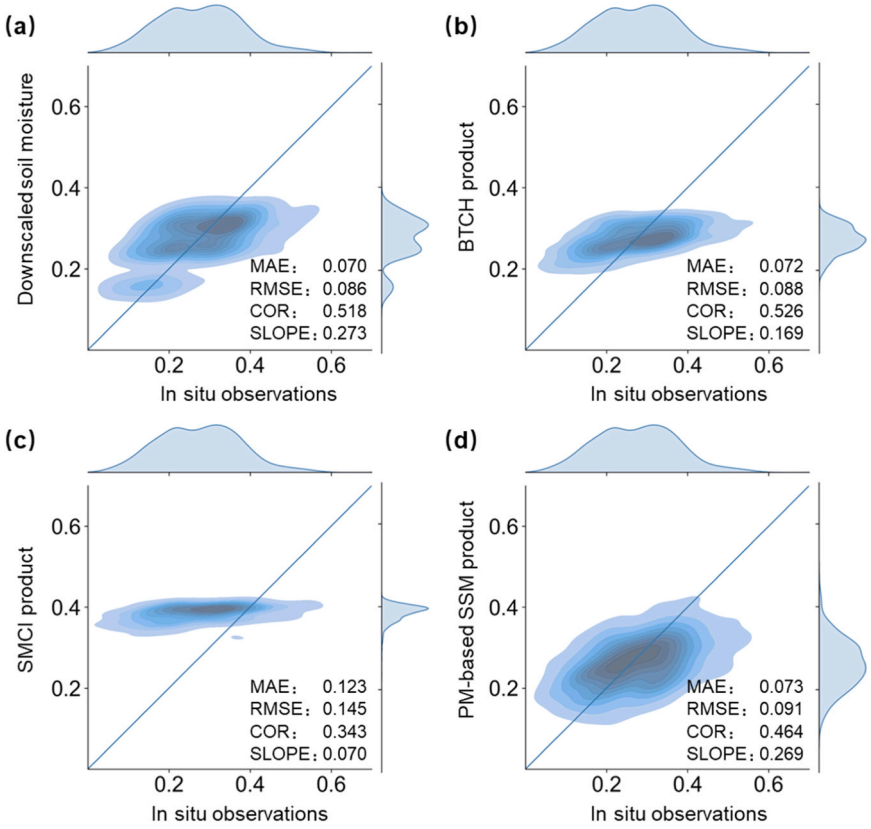


Fig. 4. Kernel density distributions and statistical comparisons of downscaled soil moisture and three available 1-km products against in situ observations. Panels show the relationships between in situ soil moisture and (a) downscaled predictions, (b) BTCH product, (c) SMCI product, and (d) PM-based product. Evaluation metrics were averaged over 20 dates from July to August during 2016–2020.

0.086 m^3/m^3. Despite some discrepancies, the downscaled soil moisture was able to reasonably capture temporal fluctuations observed in the field data, as evidenced by a Pearson correlation coefficient (COR) of 0.518. However, linear regression analysis indicated a tendency to underestimate soil moisture, with a regression slope (SLOPE) of 0.273, suggesting that the model produced lower values than observed in situ.

For comparison, the 1-km BTCH soil moisture product yielded a slightly higher correlation (COR = 0.526), but exhibited larger errors, with both MAE and RMSE exceeding those of the proposed method. In addition, the BTCH product had a lower regression slope (SLOPE = 0.169), indicating poorer agreement in terms of magnitude. The SMCI and PM-based products performed even worse, characterized by higher errors and weaker correlations. In particular, SMCI exhibited the weakest consistency with in situ observations, with a correlation coefficient as low as 0.343, suggesting a limited capability to capture the spatiotemporal variability of surface soil moisture at the observation scale.

To further assess the impact of incorporating physical constraints, we

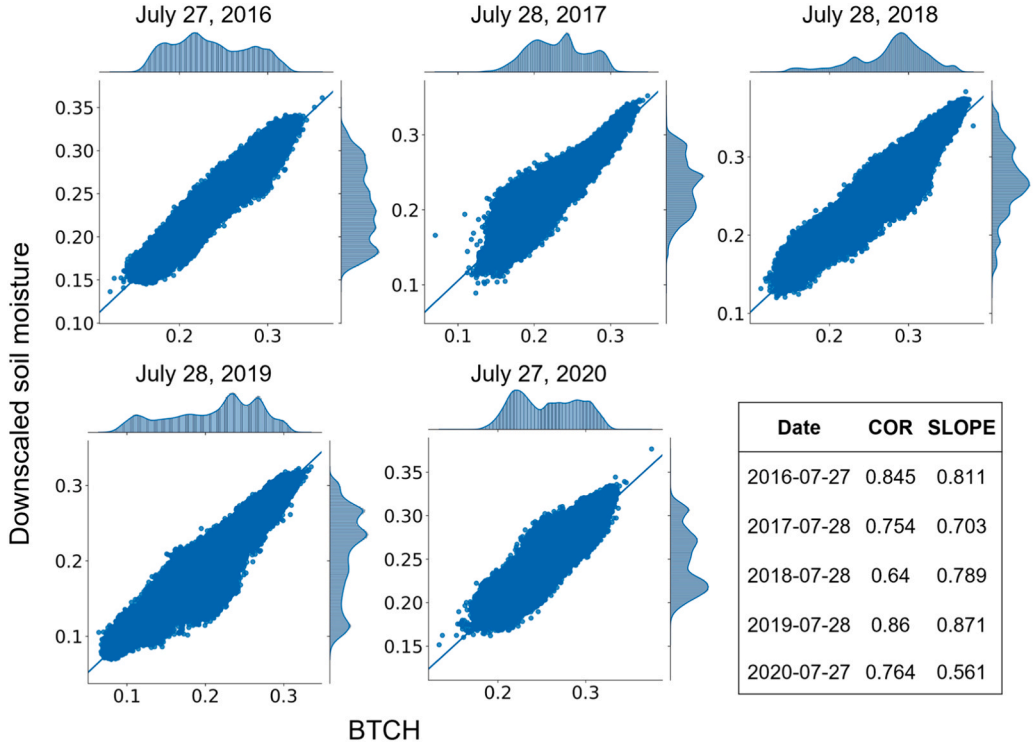
evaluated the performance of the DISPATCH-constrained downscaling method against in situ observations. Table 3 summarizes the relative change rates in four evaluation metrics compared to two baselines: the unconstrained model and the BTCH 1-km soil moisture product. Compared to the unconstrained model, the DISPATCH-constrained approach consistently improved accuracy, particularly in 2017, 2019, and 2020, with reductions in both MAE and RMSE. The most notable improvement occurred in 2019, with a 6.89 % reduction in MAE, indicating that physical constraints effectively reduced bias introduced during the coarse-to-fine downscaling process. Beyond error reduction, the constrained model also achieved better agreement in magnitude and variability with in situ observations. The regression slope (SLOPE) improved by up to 76.83 % (2019), and COR increased in most years, especially in 2017 and 2020. When benchmarked against the BTCH product, the DISPATCH-constrained model also showed lower MAE and RMSE for most years. While changes in COR were mixed, significant gains were again observed in SLOPE, suggesting the constrained model provided a more accurate representation of soil moisture magnitude dynamics.

3.3. Comparison among available 1-km soil moisture products

The performance of the DISPATCH-constrained downscaled soil moisture was further assessed against the publicly available 1-km BTCH dataset, which serves as a widely used high-resolution benchmark. While the proposed method generally enhances predictive accuracy, it may occasionally lead to a slight reduction in linear agreement with existing products like BTCH (Figs. 5–6). Comparisons were conducted for five representative dates between 2016 and 2020 to evaluate spatial consistency and magnitude alignment (Fig. 5). On 27 July 2016 and 28 July 2019, the downscaled soil moisture exhibited strong agreement with BTCH, with both the COR and SLOPE exceeding 0.8, indicating close linear alignment and magnitude consistency. However, on other dates such as 28 July 2018, COR dropped to 0.64, and on 27 July 2020, the SLOPE declined to 0.56, suggesting a narrower dynamic range in the downscaled results compared to BTCH. To further investigate the influence of the DISPATCH constraint, we compared downscaling results generated with and without the constraint over 20 dates (Fig. 6). On 14 out of 20 dates, the DISPATCH-constrained model yielded lower MAE and RMSE, indicating consistent error reduction. Notably, on 28 July 2018, the MAE and RMSE decreased by 16.2 % and 13.2 %, respectively, demonstrating the clear benefit of physically guided adjustments on that date. However, slight error increases were observed on some dates (particularly in 2019), suggesting that the constraint's effectiveness may be sensitive to specific climatic or surface conditions. Additionally, small reductions in COR and SLOPE were recorded on several dates when using the constrained model. Averaged across all dates, the declines in COR and SLOPE were modest, at 1.21 % and 0.85 %, respectively.

When evaluated relative to the BTCH reference, the proposed downscaling framework outperforms existing 1-km soil moisture products by achieving lower errors, higher correlations, and more stable and physically consistent estimates (Fig. 7). Based on the evaluation metrics (MAE, RMSE, COR, and SLOPE) for the downscaled soil moisture and existing 1-km products, the proposed downscaling framework consistently achieved the lowest MAE and RMSE values across all 20 periods. The downscaled soil moisture exhibited small temporal fluctuations, with MAE values generally below 0.03 m^3/m^3 and RMSE values below 0.04 m^3/m^3, substantially outperforming both the SMCI and PM-based products. In terms of correlation, the downscaled estimates maintained the highest COR values (mostly 0.6–0.9), demonstrating a strong agreement with the BTCH reference. In contrast, the PM-based product showed moderate correlations with larger temporal variability, while the SMCI product exhibited low correlations (< 0.4) during most periods. Regression slopes further confirmed this pattern: the proposed method-maintained slopes close to 0.7–0.9, suggesting a robust representation of magnitude consistency, whereas SMCI and PM-based showed larger deviations and occasional underestimations.

When compared with other 1-km downscaled products (Fig. 8), the proposed downscaled soil moisture exhibits stronger consistency with the PM-based product and relatively weaker agreement with the SMCI product. For the comparison with the PM-based product, most dates display favorable evaluation metric values, represented by deeper blue



Date	COR	SLOPE
2016-07-27	0.845	0.811
2017-07-28	0.754	0.703
2018-07-28	0.64	0.789
2019-07-28	0.86	0.871
2020-07-27	0.764	0.561

Fig. 5. Scatterplots between the 1-km downscaled soil moisture and the BTCH product on five selected dates.

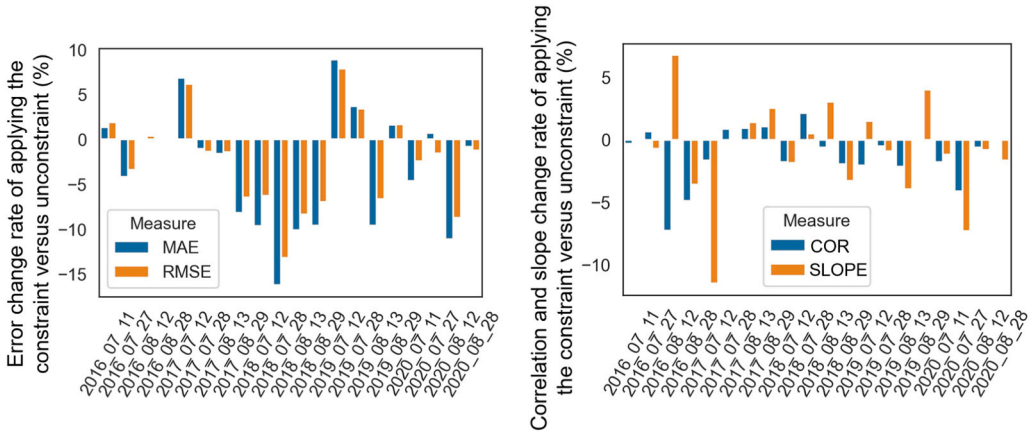


Fig. 6. Change rates in evaluation metrics between downscaled soil moisture predictions with and without the DISPATCH constraint, using the BTCH product as the reference. Negative values for MAE and RMSE indicate reduced error after applying the constraint, while positive values for COR and SLOPE indicate improved agreement in spatial pattern and magnitude.

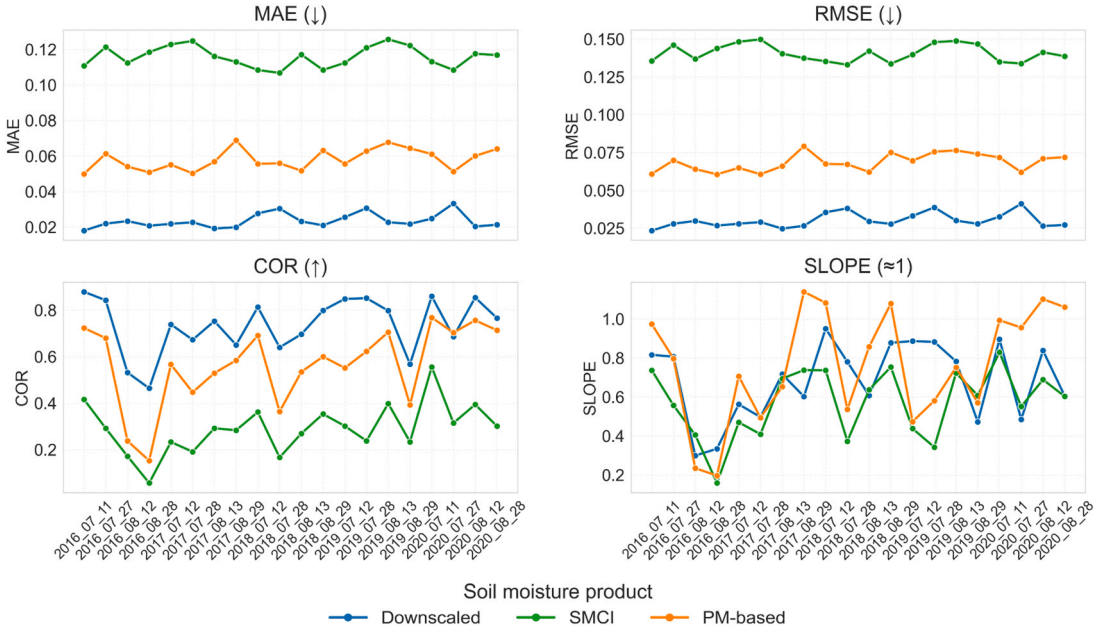


Fig. 7. Temporal variations of evaluation metrics (MAE, RMSE, COR, and SLOPE) for downscaled soil moisture and three available 1-km products relative to the BTCH reference.

or red tones in the heatmaps, which indicate smaller errors and stronger correlations. Similarly, based on the statistical distributions across 20 dates, the downscaled estimates achieve lower average errors (MAE ≈ 0.05–0.06 m^3/m^3, RMSE ≈ 0.07–0.08 m^3/m^3) and higher correlations (COR ≈ 0.5–0.7). These results demonstrate that the downscaled soil moisture effectively captures the spatiotemporal variability represented in the PM-based dataset. Although the agreement with SMCI is less pronounced, the downscaled product still maintains moderate performance across most periods.

3.4. Comparative analysis of various physical constraints

The performance of the various downscaling methods listed in Table 3 was evaluated using in situ soil moisture observations as reference data (Fig. 9). Among the baseline methods, $Base_2$ slightly outperformed $Base_1$, showing reduced errors and a higher COR value. The inclusion of soil properties provided a modest improvement, suggesting that these variables contribute to more accurate estimation of spatial soil moisture variability. The addition of physically meaningful auxiliary variables (i.e., TVDI and SEE) further enhanced model performance. Both $Variable_1$ and $Variable_2$ exhibited lower MAE and RMSE, along with higher COR and SLOPE values, compared to their respective baseline counterparts. These results confirm the value of integrating physically interpretable predictors into the trend surface model. The greatest improvements, however, were achieved by the physically constrained approaches ($Proposed_1$–$Proposed_4$). The four methods achieved the lowest errors and the highest agreement with ground measurements. In particular, $Proposed_2$, which incorporates enhanced topographic influence and reduced precipitation weighting, yielded the best overall performance across all metrics.

Notably, $Proposed_4$ also outperformed $Proposed_3$, indicating that reducing precipitation influence at finer scales improves model alignment with in situ conditions. This finding is consistent with previous research suggesting that precipitation exhibits weak correlation with point-scale soil moisture due to rapid redistribution, infiltration, and evapotranspiration processes in complex terrain. In contrast, the BTCH product achieved relatively high correlation with in situ observations but exhibited larger errors and lower SLOPE values, indicating limited

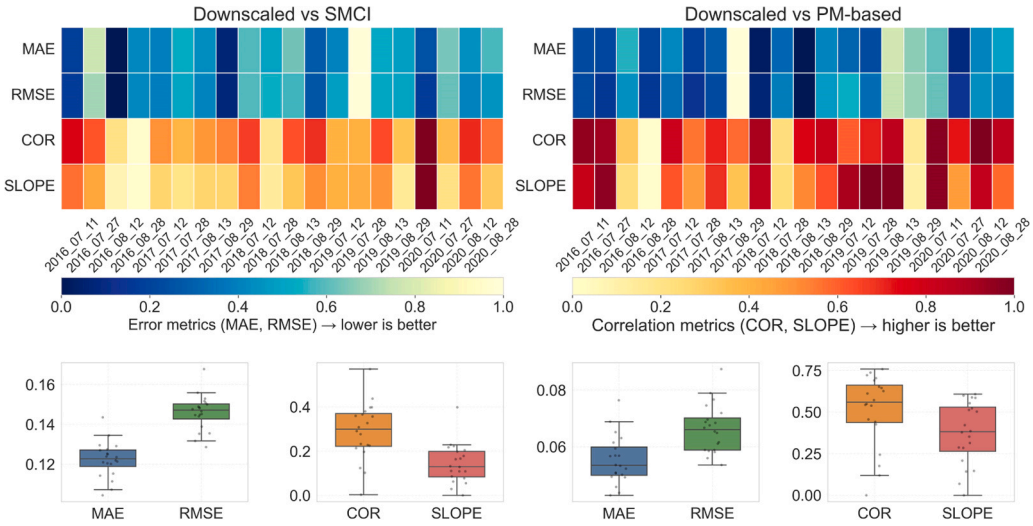


Fig. 8. Temporal patterns and statistical distributions of accuracy metrics for downscaled soil moisture relative to the SMCI and PM-based products. The top panels illustrate the temporal evolution of metrics (heatmaps), while the bottom panels show the distributions of error (MAE, RMSE) and correlation (COR, SLOPE) metrics across 20 dates.

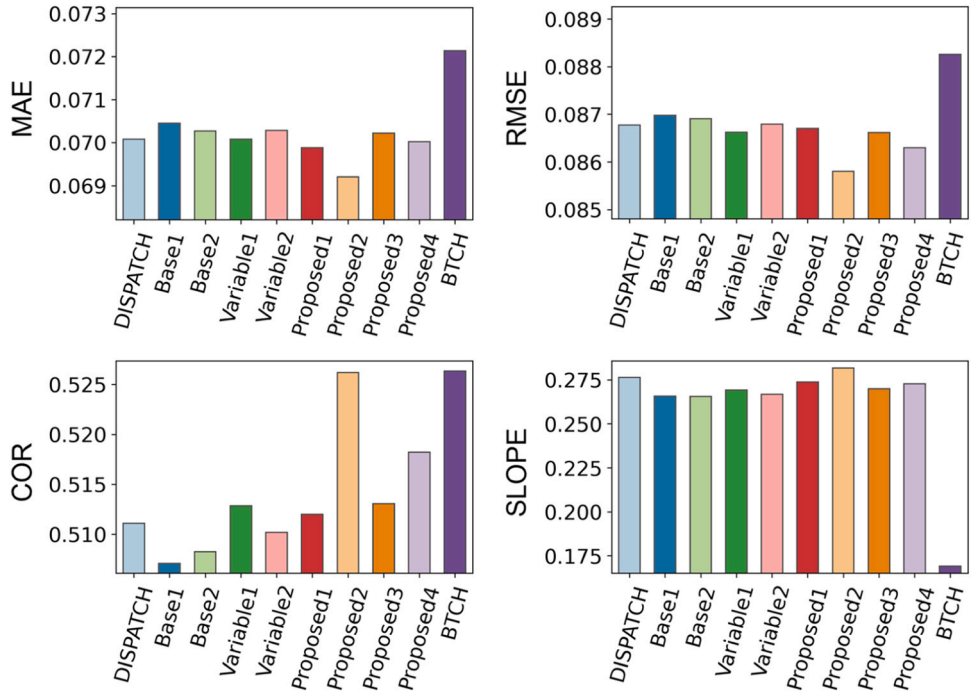


Fig. 9. Comparison of soil moisture downscaling methods based on different physical feature configurations, as defined in Table 3, using in situ observations as reference.

ability to capture soil moisture magnitude. Meanwhile, predictions from the original DISPATCH method achieved competitive MAE, RMSE, and SLOPE values but maintained only moderate correlation with ground measurements, likely due to its semi-empirical structure and assumption of moisture-limited conditions.

Further analyses were conducted using the 1-km BTCH product as a reference to compare the performance of different downscaling methods (Fig. 10). When comparing $Base_1$ and $Base_2$, the difference rates of $Base_2$ − $Base_1$ for MAE and RMSE were generally negative, indicating that $Base_2$ produced lower errors and thus improved accuracy. $Base_2$ also achieved higher COR and SLOPE values, reflecting better agreement with the BTCH product. The improvements from $Variable_1$ and $Variable_2$ over their respective baselines were less pronounced and showed greater variation across dates. Notably, introducing DISPATCH constraints to $Variable_1$ (i.e., $Proposed_1$ and $Proposed_2$) did not consistently enhance performance. On some dates, COR and SLOPE even declined compared to $Variable_1$, suggesting that the imposed constraints may not have aligned well with BTCH's underlying spatial characteristics.

In contrast, $Proposed_3$, which applied DISPATCH-based topographic weighting to $Variable_2$, achieved consistently lower MAE and RMSE across most dates. However, the introduction of precipitation constraints in $Proposed_4$ led to decreased accuracy relative to $Proposed_3$, likely due to the stronger precipitation dependency in the BTCH product at the 1-km scale. A direct comparison between $Proposed_3$ and the original DISPATCH method showed that $Proposed_3$ yielded both lower errors and a substantially higher correlation coefficient, confirming the

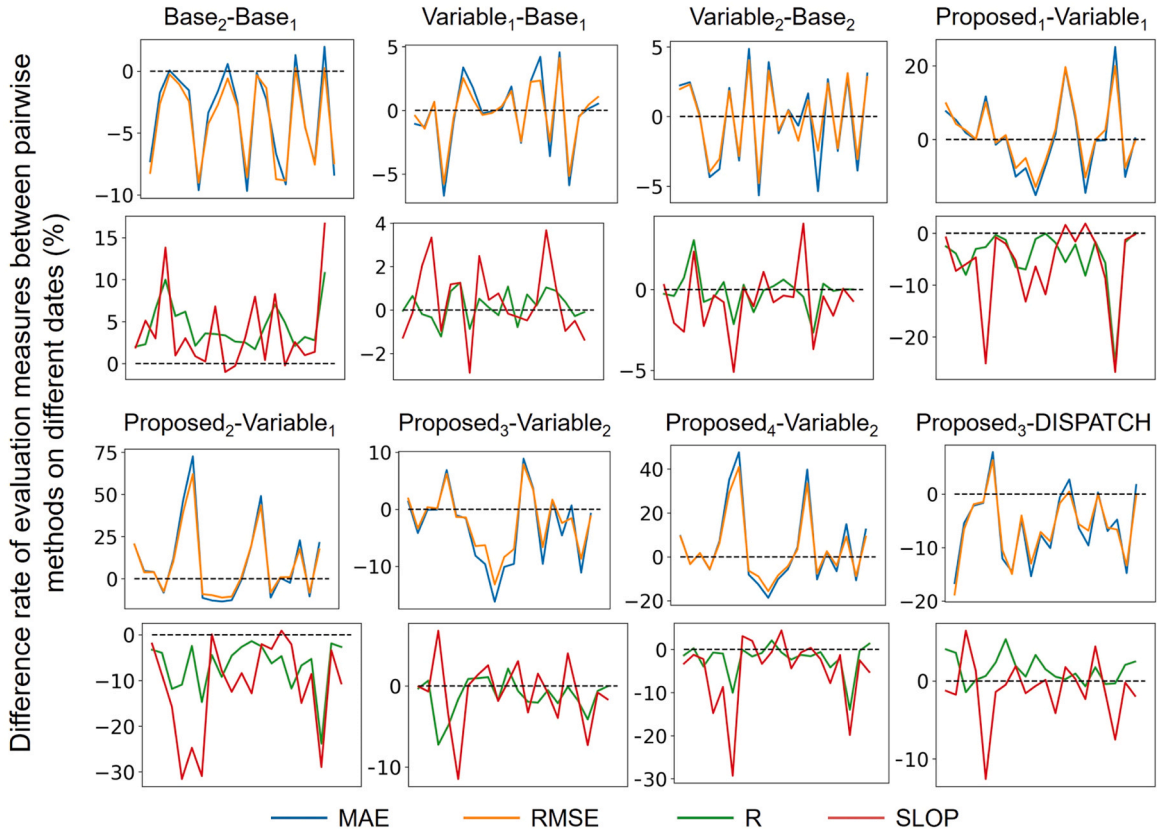


Fig. 10. Comparison of downscaling methods informed by soil moisture-related physical features, using the 1-km BTCH product as reference. The difference rate is calculated by subtracting evaluation metrics of paired methods. For MAE and RMSE, a positive value indicates increased error; for COR and SLOPE, a positive value indicates decreased performance.

advantage of integrating physical constraints within a machine learning framework when evaluated against data-driven benchmarks like BTCH.

The comparative performance of downscaling methods differed when evaluated against in situ observations versus the BTCH product, largely due to differences in spatial scale and inherent uncertainties in the BTCH downscaled data. Analysis revealed that reducing the influence of precipitation via DISPATCH constraints (i.e., Proposed$_2$ and Proposed$_4$) improved consistency with in situ measurements (Fig. 9). This is because precipitation exhibits weak correlation with soil moisture at the site scale (average correlation coefficient ~0.09), in contrast to moderate correlations (~0.5) at the 1-km and 25-km scales (Fig. 11).

However, when using the BTCH product as the reference, further reducing precipitation influence did not improve model performance. This is expected, as BTCH is trained on 1-km scale data where precipitation remains a dominant predictor. In contrast, topographic features (e.g., elevation, slope orientation) showed stronger correlations with in situ soil moisture than with gridded soil moisture products at coarser scales, indicating their greater relevance at fine resolutions. Nonetheless, since auxiliary features were extracted from 1-km data, this likely underrepresents their true correlation with point-scale soil moisture.

Most soil properties (e.g., texture, bulk density, organic carbon) exhibited the strongest correlations with the 1-km BTCH product, and incorporating them into the model (Base$_2$, Variable$_2$, Proposed$_3$, Proposed$_4$) substantially improved consistency with BTCH (Fig. 10). The inclusion of TVDI and SEE also enhanced downscaling accuracy. However, once soil properties were included, the marginal benefit of TVDI and SEE diminished, likely due to their overlapping contributions to capturing soil moisture spatial variability.

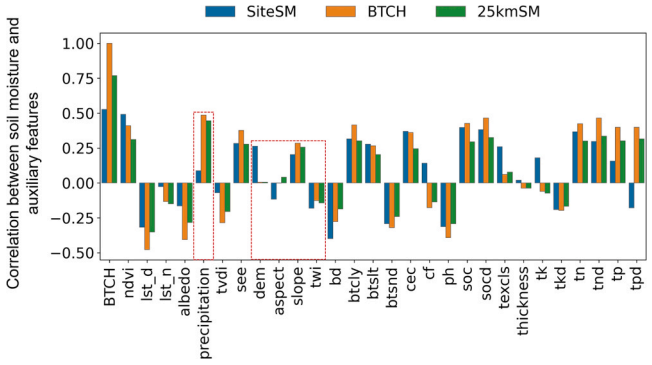


Fig. 11. Correlation coefficients between soil moisture at different spatial scales (in situ observations, 1-km BTCH, and 25-km ESA CCI) and corresponding auxiliary features.

4. Discussion

4.1. Advantages of physics-guided machine learning for soil moisture downscaling

Physics-guided machine learning has emerged as a promising direction for improving the accuracy and physical credibility of soil moisture downscaling (Meng et al., 2024). A key challenge in downscaling is that the statistical relationships learned at coarse resolution may not hold at finer scales, leading to inconsistencies in spatial patterns, scale-dependent biases, and unstable temporal dynamics. The framework developed in this study addresses this challenge by embedding physics-guided constraints into a machine learning architecture, thereby strengthening both predictive performance and scale-aware physical consistency.

The core innovation of our approach lies in the integration of physically meaningful constraints derived from the DISPATCH paradigm into a flexible data-driven model. Instead of using DISPATCH purely as an independent downscaling algorithm, we incorporate its mechanistic principles, such as the evaporative efficiency constraint and thermal–moisture coupling, into the model training process. Across extensive evaluations, the proposed framework consistently outperformed the three representative 1-km soil moisture datasets used as benchmarks (Sections 3.2–3.3). Relative to BTCH (regional machine learning ensemble), PM-based SSM (physics-guided), and SMCI (machine learning-based nationwide product), our method achieved the lowest MAE and RMSE values and the highest correlation with in situ measurements. These results demonstrate that the hybrid framework effectively combines the strengths of physical and statistical approaches, producing downscaled soil moisture estimates that better capture both spatial gradients and temporal variability over the Tibetan Plateau. The balanced comparison with these complementary baselines highlights the robustness and methodological advantage of the proposed strategy.

Compared with purely physical models, such as the semi-empirical DISPATCH algorithm, our approach exhibits stronger adaptability to heterogeneous terrain, vegetation mosaics, and complex climatic regimes. DISPATCH is known to perform well in semi-arid or moisture-limited regions, but its assumptions—particularly regarding evaporative efficiency and surface temperature–moisture coupling—are less valid in areas with strong land surface heterogeneity or seasonal hydrological extremes (Ojha et al., 2021; Zheng et al., 2021). This limitation is evident in our study area, where standalone DISPATCH shows larger prediction errors and weaker spatial coherence. In contrast, the physics-guided machine learning framework retains the physical strengths of DISPATCH while substantially reducing its structural biases. As seen in Figs. 9–10, the proposed approach achieved lower errors, higher spatial agreement, and stronger temporal consistency, illustrating the synergy between physical constraints and machine learning flexibility.

The strength of this hybrid approach lies in combining the complementary advantages of physical and data-driven models (Chen et al., 2023). Physical models embed mechanistic understanding of soil–vegetation–atmosphere interactions, ensuring that predictions are consistent with environmental laws. However, they are often constrained by simplifying assumptions, incomplete parameterization, and limited adaptability to diverse conditions. In contrast, data-driven machine learning models can flexibly capture complex, nonlinear relationships from multi-source datasets, including ground observations, remote sensing products, numerical simulations, and even social-sensing data. Without physical constraints, however, machine learning models may generate outputs that violate established environmental principles, particularly when extrapolating beyond the training domain (Chen et al., 2020). By embedding DISPATCH-informed constraints into the learning process, our framework bridges this gap, yielding predictions that are accurate and robust under diverse environmental conditions.

4.2. Scale-dependent variability in auxiliary feature importance

Overcoming the limitations of scale-invariant assumptions requires the development of cross-scale models that can capture spatial heterogeneity and nonlinear relationships (Meng et al., 2024). Understanding how soil moisture interacts with auxiliary features across multiple spatial scales is essential for building physically informed, statistical–mechanistic hybrid models. Although significant theoretical progress has been made in scale-effect analysis and scale conversion, producing high-resolution products that meet the needs of precision agriculture and fine-scale hydrological modeling still demands methodological innovation (Ge et al., 2019; Wu and Li, 2009). Our results demonstrate that the influence of auxiliary features on soil moisture prediction is inherently scale-sensitive, and therefore should be modeled adaptively rather than through fixed-weight relationships.

In this study, soil properties (such as texture, bulk density, and soil organic carbon) were found to be stable and highly effective predictors, reflecting their fundamental role in controlling soil water retention and transmission (Rasheed et al., 2022). Physically derived indices, including the TVDI and SEE, contributed additional information by capturing short-term variability associated with vegetation water stress and surface energy fluxes (Ali et al., 2024; Lehmann et al., 2018). However, when these indices were used together with soil properties, the performance gains plateaued, suggesting redundancy once the dominant physical controls were adequately represented (Figs. 9–10).

A key insight from our analysis is that the relative importance of physical drivers changes with spatial scale. Precipitation, for example, showed weak correlation with point-scale soil moisture but became markedly more influential at aggregated resolutions (1 km and 25 km), where spatial averaging reduces localized noise (Fig. 11). Conversely, topographic factors such as elevation and slope exhibited stronger correlations with in situ soil moisture than with gridded products, underscoring their importance at fine resolutions where terrain governs hydrological redistribution and retention patterns. These findings highlight a central challenge in downscaling: no single predictor maintains consistent importance across scales, and static weighting schemes are insufficient to capture such complexity (Zhang et al., 2025). By comparing the coarse-scale machine learning-predicted trend surfaces with physics-based downscaling results, we dynamically adjusted the influence of key predictors (i.e., precipitation and topography) at finer resolutions. However, it also revealed trade-offs: reducing the weight of precipitation enhanced agreement with in situ measurements but reduced consistency with the benchmark BTCH product. Such results suggest that uniform constraint strategies may be suboptimal, and future downscaling methods should adopt adaptive, driver-specific adjustments that respond to both spatial context and the scale of the reference product (Ge et al., 2019).

4.3. Limitations and future works

While the proposed framework shows strong potential, several limitations remain. First, the analysis period was restricted to summer months (July–August), limiting its applicability to other seasons with different hydrometeorological regimes such as snowmelt, frozen soils, or rainfall–runoff dominance. Second, validation was conducted in a single region, leaving the model's transferability to other climates, vegetation types, and land covers untested. Third, although the DISPATCH-based constraints provided a physics-informed adjustment, their implementation here was heuristic, lacking a fully quantitative optimization procedure. Moreover, systematic characterization of scale effects across diverse environmental gradients remains an open challenge for soil moisture downscaling research (Markham et al., 2023). Despite these limitations, this study addresses two major gaps in current downscaling practices: integrating physics-guided machine learning and explicitly modeling scale-sensitive feature behavior. Future research should focus on developing generalizable constraint-learning schemes that adapt to different environmental contexts, expanding seasonal and spatial validation coverage, and testing model robustness in diverse landscapes. Additionally, ensemble or hybrid approaches that combine multiple physical models, machine learning architectures, and data assimilation techniques could further improve predictive skill and better capture the dynamic nature of soil moisture variability across scales (Meng et al., 2024).

5. Conclusions

This study proposed a physics-guided machine learning framework for soil moisture downscaling, integrating DISPATCH-based constraints to adaptively adjust feature influence across spatial scales. This approach improves both the spatial resolution and physical consistency of soil moisture products, offering robust performance in heterogeneous

environments. Our results support a recommended paradigm for downscaling: combine data-driven flexibility with physically informed constraints through adaptive, scale-aware feature weighting, enabling accurate and transferable high-resolution products for hydrological, agricultural, and climate applications.

CRediT authorship contribution statement

Yong Ge: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Jin Chen:** Writing – review & editing, Conceptualization. **Mo Zhang:** Writing – review & editing, Investigation, Conceptualization. **Die Zhang:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Ali, A.M., Lü, H., Zhu, Y., Ahmed, K., Farhan, M., Qasim, M., 2024. Spatio-temporal remote sensing evaluation of drought impact on vegetation dynamics in Balochistan, Pakistan. Theor. Appl. Clim. 155, 8611–8632. https://doi.org/10.1007/s00704-024-05148-6.

Brocca, L., Hasenauer, S., Lacava, T., Melone, F., Moramarco, T., Wagner, W., Dorigo, W., Matgen, P., Martínez-Fernández, J., Llorens, P., Latron, J., Martin, C., Bittelli, M., 2011. Soil moisture estimation through ASCAT and AMSR-E sensors: an intercomparison and validation study across Europe. Remote Sens Environ. 115, 3390–3408. https://doi.org/10.1016/j.rse.2011.08.003.

Chen, Q., Miao, F., Xu, Z.-X., Wang, H., Yang, L., Tang, Z., 2019. Downscaling of remote sensing soil moisture products based on TVDI in complex terrain areas, in: 2019. Int. Conf. Meteorol. Obs. (ICMO) 1–3. https://doi.org/10.1109/ICMO49322.2019.9026012.

Chen, B., Wang, Y., Wang, R., Zhu, Z., Ma, L., Qiu, X., Dai, W., 2020. The gray-box based modeling approach integrating both mechanism-model and data-model: the case of atmospheric contaminant dispersion. Symmetry 12. https://doi.org/10.3390/sym12020254.

Chen, Y., Yang, K., Qin, J., Jiang, Y., La, Z., Zhou, X., Tian, J., Zhao Long, 2022. The multiscale observation network of soil temperature and moisture on the central Tibetan Plateau (2010-2021). Natl. Tibet. Plateau Data Cent. https://doi.org/10.11888/Terre.tpdc.271918.

Chen, C., Zhang, H., Shi, W., Zhang, W., Xue, Y., 2023. A novel paradigm for integrating physics-based numerical and machine learning models: A case study of eco-hydrological model. Environ. Model. Softw. 163, 105669. https://doi.org/10.1016/j.envsoft.2023.105669.

Cheng, Y., Liu, H., 2024. The crucial role of soil moisture in the evolution of forest cover in Asia since the Last Glacial Maximum. Innovation 5, 100594. https://doi.org/10.1016/j.xinn.2024.100594.

Crow, W.T., Berg, A.A., Cosh, M.H., Loew, A., Mohanty, B.P., Panciera, R., de Rosnay, P., Ryu, D., Walker, J.P., 2012. Upscaling sparse ground-based soil moisture observations for the validation of coarse-resolution satellite soil moisture products. Rev. Geophys. 50. https://doi.org/10.1029/2011RG000372.

Dos Santos Araujo, D.C., Gico Lima Montenegro, S.M., Ribeiro Neto, A., da Silva, S.F., 2024. Evaluation of satellite-based soil moisture for agricultural drought monitoring in the Brazilian semiarid region. Remote Sens Appl. 33, 101111. https://doi.org/10.1016/j.rsase.2023.101111.

Ge, Y., Jin, Y., Stein, A., Chen, Y., Wang, Jianghao, Wang, Jinfeng, Cheng, Q., Bai, H., Liu, M., Atkinson, P.M., 2019. Principles and methods of scaling geospatial Earth science data. Earth Sci. Rev. 197, 102897. https://doi.org/10.1016/j.earscirev.2019.102897.

Jin, Y., Ge, Y., Liu, Y., Chen, Y., Zhang, H., Heuvelink, G.B.M., 2021. A Machine Learning-Based Geostatistical Downscaling Method for Coarse-Resolution Soil Moisture Products. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens 14, 1025–1037. https://doi.org/10.1109/JSTARS.2020.3035386.

Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., 2021. Physics-informed machine learning. Nat. Rev. Phys. 3, 422–440. https://doi.org/10.1038/s42254-021-00314-5.

Lehmann, P., Merlin, O., Gentine, P., Or, D., 2018. Soil Texture Effects on Surface Resistance to Bare-Soil Evaporation, 405 Geophys Res Lett. 45 (10), 310–398. https://doi.org/10.1029/2018GL078803.

Li, L., Dai, Y., Wei, Z., Shangguan, W., Wei, N., Zhang, Y., Li, Q., Li, X.-X., 2024. Enhancing Deep Learning Soil Moisture Forecasting Models by Integrating Physics-based Models. Adv. Atmos. Sci. 41, 1326–1341. https://doi.org/10.1007/s00376-023-3181-8.

Li, Q., Shi, G., Shangguan, W., Nourani, V., Li, J., Li, L., Huang, F., Zhang, Y., Wang, C., Wang, D., Qiu, J., Lu, X., Dai, Y., 2022. A 1km daily soil moisture dataset over China using in situ measurement and machine learning. Earth Syst. Sci. Data 14, 5267–5286. https://doi.org/10.5194/essd-14-5267-2022.

Liu, Y., Jing, W., Wang, Q., Xia, X., 2020. Generating high-resolution daily soil moisture by using spatial downscaling techniques: a comparison of six machine learning algorithms. Adv. Water Resour. 141, 103601. https://doi.org/10.1016/j.advwatres.2020.103601.

Markham, K., Frazier, A.E., Singh, K.K., Madden, M., 2023. A review of methods for scaling remotely sensed data for spatial pattern analysis. Land. Ecol. 38, 619–635. https://doi.org/10.1007/s10980-022-01449-1.

Meng, C., Griesemer, S., Cao, D., Seo, S., Liu, Y., 2025. When physics meets machine learning: a survey of physics-informed machine learning. Mach. Learn. Comput. Sci. Eng. 1, 20. https://doi.org/10.1007/s44379-025-00016-0.

Meng, X., Zeng, J., Yang, Y., Zhao, W., Ma, H., Letu, H., Zhu, Q., Liu, Y., Wang, P., Peng, J., 2024. High-resolution soil moisture mapping through passive microwave remote sensing downscaling. Innov. Geosci. 2, 100105. https://doi.org/10.59717/j.xinn-geo.2024.100105.

Merlin, O., Rudiger, C., Bitar, A.A., Richaume, P., Walker, J.P., Kerr, Y.H., 2012. Disaggregation of SMOS Soil Moisture in Southeastern Australia. IEEE Trans. Geosci. Remote Sens 50, 1556–1571. https://doi.org/10.1109/TGRS.2011.2175000.

Mohanty, B.P., Cosh, M.H., Lakshmi, V., Montzka, C., 2017. Soil moisture remote sensing: state-of-the-science. Vadose Zone J. 16, vzj2016.10.0105. https://doi.org/10.2136/vzj2016.10.0105.

Molero, B., Merlin, O., Malbéteau, Y., Al Bitar, A., Cabot, F., Stefan, V., Kerr, Y., Bacon, S., Cosh, M.H., Bindlish, R., Jackson, T.J., 2016. SMOS disaggregated soil moisture product at 1km resolution: Processor overview and first validation results. Remote Sens Environ. 180, 361–376. https://doi.org/10.1016/j.rse.2016.02.045.

Ojha, N., Merlin, O., Suere, C., Escorihuela, M.J., 2021. Extending the Spatio-Temporal Applicability of DISPATCH Soil Moisture Downscaling Algorithm: A Study Case Using SMAP, MODIS and Sentinel-3 Data. Front Environ. Sci. 9, 2021. https://doi.org/10.3389/fenvs.2021.555216.

Peng, J., Loew, A., Merlin, O., Verhoest, N.E.C., 2017. A review of spatial downscaling of satellite remotely sensed soil moisture. Rev. Geophys. 55, 341–366. https://doi.org/10.1002/2016RG000543.

Rasheed, M.W., Tang, J., Sarwar, A., Shah, S., Saddique, N., Khan, M.U., Imran Khan, M., Nawaz, S., Shamshiri, R.R., Aziz, M., Sultan, M., 2022. Soil moisture measuring techniques and factors affecting the moisture dynamics: a comprehensive review. Sustainability 14. https://doi.org/10.3390/su141811538.

Senanayake, I.P., Pathira Arachchilage, K.R.L., Yeo, I.-Y., Khaki, M., Han, S.-C., Dahlhaus, P.G., 2024. Spatial downscaling of satellite-based soil moisture products using machine learning techniques: a review. Remote Sens 16. https://doi.org/10.3390/rs16122067.

Seneviratne, S.I., Corti, T., Davin, E.L., Hirschi, M., Jaeger, E.B., Lehner, I., Orlowsky, B., Teuling, A.J., 2010. Investigating soil moisture–climate interactions in a changing climate: a review. Earth Sci. Rev. 99, 125–161. https://doi.org/10.1016/j.earscirev.2010.02.004.

Shangguan, W., Li, Q., Shi, G., 2024. A 1 km daily soil moisture dataset over China based on in-situ measurement (2000-2022). Natl. Tibet. Plateau Data Cent. https://doi.org/10.11888/Terre.tpdc.272415.

Shangguan, Y., Min, X., Shi, Z., 2023a. Inter-comparison and integration of different soil moisture downscaling methods over the Qinghai-Tibet Plateau. J. Hydrol. (Amst.) 617, 129014. https://doi.org/10.1016/j.jhydrol.2022.129014.

Shangguan, Y., Shi, Z., Min, X., 2023b. A 1 km daily soil moisture dataset over the Qinghai-Tibet Plateau (2001-2020). Natl. Tibet. Plateau Data Cent. https://doi.org/10.11888/Terre.tpdc.300224.

Song, C., Hu, G., Wang, Y., Qu, X., 2022. Downscaling ESA CCI soil moisture based on soil and vegetation component temperatures derived from MODIS data. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens 15, 2175–2184. https://doi.org/10.1109/JSTARS.2022.3155463.

Song, P., Zhang, Y., Guo, J., Shi, J., Zhao, T., Tong, B., 2022. A 1km daily surface soil moisture dataset of enhanced coverage under all-weather conditions over China in 2003–2019. Earth Syst. Sci. Data 14, 2613–2637. https://doi.org/10.5194/essd-14-2613-2022.

Song, P., Zhang, Y., Yao, P., Zhao, T., 2025. Daily all weather surface soil moisture data set with 1 km resolution in China (2003-2024). Natl. Tibet. Plateau Data Cent. https://doi.org/10.11888/Hydro.tpdc.271762.

Sun, H., Gao, J., 2023. A pixel-wise calculation of soil evaporative efficiency with thermal/optical remote sensing and meteorological reanalysis data for downscaling

microwave soil moisture. Agric. Water Manag 276, 108063. https://doi.org/10.1016/j.agwat.2022.108063.
Wang, Y., Shi, H., Yang, X., Jiang, Y., Wu, Y., Shui, J., Liu, Y., Guo, M., Li, L., 2024. Spatial downscaling of SMAP soil moisture to high resolution using machine learning over China's Loess Plateau. Catena 247, 108492. https://doi.org/10.1016/j.catena.2024.108492.
Wigneron, J.-P., Li, X., Frappart, F., Fan, L., Al-Yaari, A., De Lannoy, G., Liu, X., Wang, M., Le Masson, E., Moisy, C., 2021. SMOS-IC data record of soil moisture and L-VOD: historical development, applications and perspectives. Remote Sens Environ. 254, 112238. https://doi.org/10.1016/j.rse.2020.112238.
Wu, H., Li, Z.-L., 2009. Scale Issues in Remote Sensing: A Review on Analysis, Processing and Modeling. Sensors 9, 1768–1793. https://doi.org/10.3390/s90301768.
Xie, Q., Jia, L., Menenti, M., Chen, Q., Bi, J., Chen, Y., Wang, C., Yu, X., 2024. Evaluation of remote sensing soil moisture data products with a new approach to analyse footprint mismatch with in-situ measurements. Int J. Digit Earth 17, 2437051. https://doi.org/10.1080/17538947.2024.2437051.
Xu, Z., Sun, H., Gao, J., Wang, Y., Wu, D., Zhang, T., Xu, H., 2024. PhySoilNet: A deep learning downscaling model for microwave satellite soil moisture with physical rule constraint. Int. J. Appl. Earth Obs. Geoinf. 135, 104290. https://doi.org/10.1016/j.jag.2024.104290.
Yu, T., Guli·Jiapaer, Bao, A., Zhang, J., Tu, H., Chen, B., De Maeyer, P., Van de Voorde, T., 2023. Evaluating surface soil moisture characteristics and the performance of remote sensing and analytical products in Central Asia. J. Hydrol. 617, 128921. https://doi.org/10.1016/j.jhydrol.2022.128921.
Zhang, Q., Yuan, Q., Li, J., Wang, Y., Sun, F., Zhang, L., 2021. Generating seamless global daily AMSR2 soil moisture (SGD-SM) long-term products for the years 2013–2019. Earth Syst. Sci. Data 13, 1385–1401. https://doi.org/10.5194/essd-13-1385-2021.
Zhang, M., Zhang, D., Jin, Y., Wan, X., Ge, Y., 2025. Evolution of soil moisture mapping from statistical models to integrated mechanistic and geoscience-aware approaches. Inf. Geogr. 1, 100005. https://doi.org/10.1016/j.infgeo.2025.100005.
Zhao, H., Li, J., Yuan, Q., Lin, L., Yue, L., Xu, H., 2022. Downscaling of soil moisture products using deep learning: Comparison and analysis on Tibetan Plateau. J. Hydrol. 607, 127570. https://doi.org/10.1016/j.jhydrol.2022.127570.
Zhao, W., Wen, F., Wang, Q., Sanchez, N., Piles, M., 2021. Seamless downscaling of the ESA CCI soil moisture data at the daily scale with MODIS land products. J. Hydrol. 603, 126930. https://doi.org/10.1016/j.jhydrol.2021.126930.
Zheng, J., Lü, H., Crow, W.T., Zhao, T., Merlin, O., Rodriguez-Fernandez, N., Shi, J., Zhu, Y., Su, J., Kang, C.S., Wang, X., Gou, Q., 2021. Soil moisture downscaling using multiple modes of the DISPATCH algorithm in a semi-humid/humid region. Int. J. Appl. Earth Obs. Geoinf. 104, 102530. https://doi.org/10.1016/j.jag.2021.102530.
Zhu, Z., Bo, Y., Sun, T., 2023. Spatial downscaling of satellite soil moisture products based on apparent thermal inertia: considering the effect of vegetation condition. J. Hydrol. (Amst.) 616, 128824. https://doi.org/10.1016/j.jhydrol.2022.128824.